

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Semester Final  
Duration: 1 hour 50 min

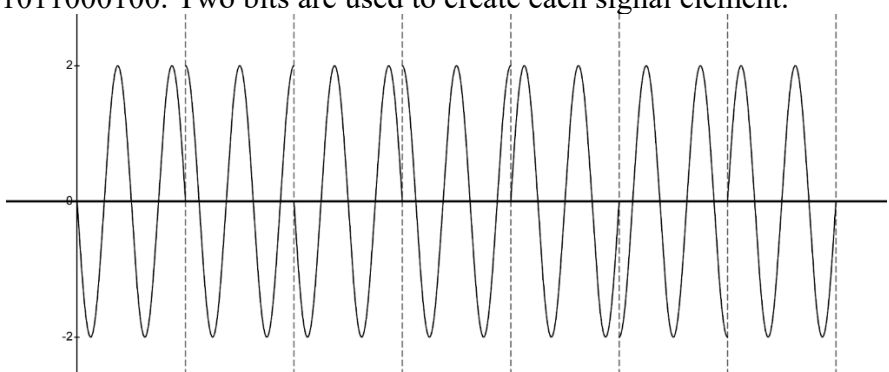
Semester: Summer 2025  
Full Marks: 60

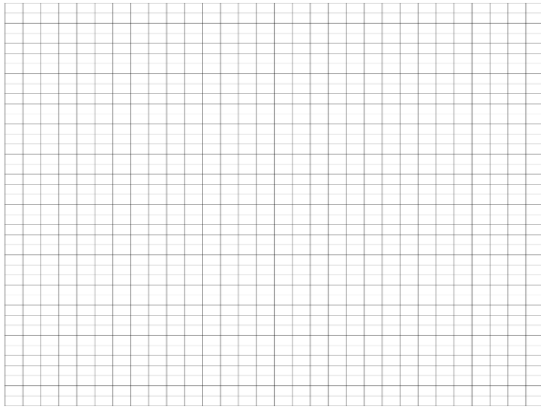
**CSE 320/EEE361/ECE361: Data Communications**

Answer the following questions.  
Figures in the right margin indicate marks.

**SET A**

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
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| <b>1.</b><br><b>[CO2]</b> | a)                                                                                                                                                                                                                                                                   | <p>In multilevel FSK, for each signal element, we want to send 2 bits at a time. We have used a carrier signal that has an amplitude of 10v and phase is 180 degrees. If the frequency changes according to the following table, <b>draw</b> the modulated signal for the bit sequence <b>0010011110</b>.</p> <table border="1" style="margin-left: auto; margin-right: auto;"><thead><tr><th>Bits</th><th>No. of cycles of the signal elements</th></tr></thead><tbody><tr><td>00</td><td>1</td></tr><tr><td>01</td><td>3</td></tr><tr><td>10</td><td>2</td></tr><tr><td>11</td><td>4</td></tr></tbody></table> <div style="text-align: center;"></div> | Bits | No. of cycles of the signal elements | 00 | 1 | 01 | 3 | 10 | 2 | 11 | 4 | [5] |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------|----|---|----|---|----|---|----|---|-----|
|                           | Bits                                                                                                                                                                                                                                                                 | No. of cycles of the signal elements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |                                      |    |   |    |   |    |   |    |   |     |
| 00                        | 1                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                      |    |   |    |   |    |   |    |   |     |
| 01                        | 3                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                      |    |   |    |   |    |   |    |   |     |
| 10                        | 2                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                      |    |   |    |   |    |   |    |   |     |
| 11                        | 4                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                      |    |   |    |   |    |   |    |   |     |
| b)                        | <p>The following analog signal has been received for the bit stream 10111011000100. Two bits are used to create each signal element.</p> <div style="text-align: center;"></div> | [1+4]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |                                      |    |   |    |   |    |   |    |   |     |

|             |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
|-------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             |    | <ul style="list-style-type: none"> <li>Which type of shift keying is used here?</li> <li><b>Draw</b> the constellation diagram for the above shift keying technique.</li> </ul>                                                                                                                                                                                                                                                                                                                           |       |
| 2.<br>[CO3] | a) | <p>FOUR analog channels are to be multiplexed using <b>FDM</b>. Required frequency bandwidths for each channel are 20 kHz, 25 kHz, 35 kHz, and 50 kHz respectively. <b>A guard band is required between every two adjacent channels. The size of each guard band is 40% of the difference between the bandwidths of those two channels.</b></p> <ul style="list-style-type: none"> <li><b>Find</b> the total bandwidth required in the output channel.</li> <li><b>Show</b> the FDM scheme in frequency domain representation assuming the starting frequency to be 100 kHz.</li> </ul>     | [4+2] |
|             | b) | <ol style="list-style-type: none"> <li>Mention some benefits of Synchronous TDM technique.</li> <li>Consider a high-performance computing system with 100 devices, but only 60% devices are active at any given time. Which TDM technique should be used for this system, and why?</li> </ol>                                                                                                                                                                                                                                                                                               | [2+2] |
|             | c) | <p>A system is using Time Division Multiplexing (TDM) to combine 3 channels together. Channel 1 is transmitting 18 kb (kilobits) in 3 seconds, channel 2 is transmitting 18 kb in 2 seconds and channel 3 is transmitting 12 kb in 1 second. If 5 bits from each channel and extra 3 synchronization bits are used to create one frame, Answer the following questions:</p> <ol style="list-style-type: none"> <li>How many input channels are being used for final TDM?</li> <li>What is the frame size</li> <li>What is the frame rate?</li> <li>What is the output data rate?</li> </ol> | [8]   |

| 3.<br>[CO4] | a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Suppose you are given the following CRC <b>Codeword 1010001</b> , where the first <b>4 bits</b> are the original <b>Dataword</b> . <ul style="list-style-type: none"><li>How many 0's are augmented at the sender side?</li><li>You are given two possible divisors: <b>1101</b> and <b>1001</b>. <b>Determine</b> which one is the correct divisor.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                      | [1+4+1]  |          |    |         |    |         |    |         |    |   |           |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----|---------|----|---------|----|---------|----|---|-----------|
|             | b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Observe the Block Coding scheme below. <table border="1"><thead><tr><th>Dataword</th><th>Codeword</th></tr></thead><tbody><tr><td>00</td><td>0010101</td></tr><tr><td>01</td><td>0101001</td></tr><tr><td>10</td><td>1000111</td></tr><tr><td>11</td><td>?</td></tr></tbody></table> <ul style="list-style-type: none"><li>If XORing the second and third codeword produces the fourth codeword, <b>find</b> the fourth codeword.</li><li><b>Calculate</b> the total number of codewords which are invalid.</li><li>How many errors can be detected and corrected in the above scheme?</li><li>Suppose the codeword received is 1010111. Is there any error in this received codeword? If there is indeed an error, <b>find</b> the original Dataword. <b>Show</b> necessary calculation.</li></ul> | Dataword | Codeword | 00 | 0010101 | 01 | 0101001 | 10 | 1000111 | 11 | ? | [1+1+4+3] |
|             | Dataword                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Codeword                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |          |    |         |    |         |    |         |    |   |           |
| 00          | 0010101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |          |    |         |    |         |    |         |    |   |           |
| 01          | 0101001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |          |    |         |    |         |    |         |    |   |           |
| 10          | 1000111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |          |    |         |    |         |    |         |    |   |           |
| 11          | ?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |          |    |         |    |         |    |         |    |   |           |
| c)          | Assume a packet is made only of four 8-bit words $(3F)_{16}$ , $(AB)_{16}$ , $(C5)_{16}$ , and $(E0)_{16}$ . <ul style="list-style-type: none"><li><b>Show</b> the checksum at the sender.</li><li>If the first data item is changed to <math>(3E)_{16}</math> and the fourth data item is changed to <math>(E1)_{16}</math> during transmission, <b>check</b> if the receiver can detect any error in this case.</li><li><b>Explain</b> the reasons of the receiver's error detection state in (ii)</li></ul> | [2+2+1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |          |    |         |    |         |    |         |    |   |           |
| 4.<br>[CO4] | a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | In a warehouse, autonomous delivery robots send status messages to a central monitoring system using the Pure ALOHA protocol. <p>Robot R1 begins transmitting a message at <math>t_1 = 12</math> ms and the transmission takes 2.5 ms. The first bit of the message reaches the monitoring system at 12.7 ms.</p> <ul style="list-style-type: none"><li><b>Determine</b> the vulnerable period for R1 along with the time window. <b>Justify</b> whether a collision will occur if another robot, R2, starts transmitting at <math>t_2 = 15</math> ms.</li></ul>                                                                                                                                                                                                                                    | [3]      |          |    |         |    |         |    |         |    |   |           |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |
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|  | <p>b) A shared bus channel uses <b>1-persistent CSMA</b>. The bus length is 2 km; signal propagation speed is <math>2 \times 10^8</math> m/s and the bandwidth of the channel is 10 Mbps.</p> <ol style="list-style-type: none"> <li><b>Calculate</b> the propagation delay across the channel. <b>Calculate</b> the vulnerable time for this scenario.</li> <li>A station senses the channel idle and begins transmitting a 2000-bits frame at <math>t = 20</math> ms. <b>Determine</b> when another station located at the farthest end of the bus will receive the last bit of the frame.</li> <li>If the network uses p-persistent CSMA instead, explain whether the possibility of collision reduces or not.</li> <li>In <b>CSMA/CD</b>, what happens if a station detects a collision?</li> </ol> | <p>[2+3<br/>+2+2<br/>]</p> |
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